SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name

Cupros 50 WP

1.2 Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses of the substance or mixture
Plant protection product

Uses advised against
No data available.

1.3 Details of the supplier of the safety data sheet

Address
Cosaco GmbH
Singapurstrasse 1
20457 Hamburg
Germany

Telephone no. +49 (0)40-2365-20
Fax no. +49 (0)40-23652-255
e-mail sds@cosaco.com

Advice on Safety Data Sheet
sdb_info@umco.de

1.4 Emergency telephone number

Americas +1 866 519 4752* (US, Canada, Mexico) *Toll Free Access
+1-760-476-3962
Europe +1-760-476-3961
Asia Pacific +1-760-476-3960
Middle East/Africa +1-760-476-3959

Global Response Access Code: 334018
Account: 14537

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification in accordance with Regulation (EC) No 1272/2008 (CLP)
Acute Tox. 4; H302
Aquatic Acute 1; H400
Aquatic Chronic 1; H410
Eye Dam. 1; H318

Classification information
Classification and labelling are based on toxicological studies performed on the product (mixture).
Classification and labelling with respect to water pollution risks are based on ecotoxicological studies performed on the product (mixture).
This product is assessed and classified using the methods and criteria below referred to in Article 9 of Regulation (EC) n° 1272/2008:
Physical hazards: determined through assessment data based on the methods or standards referred to in part 2 of Annex I to CLP
Health hazards and environmental hazards: determined through toxicological and ecotoxicological assessment data based on the methods or standards referred to in Part 3, 4 and 5 of Annex I to CLP.

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 (CLP Regulation)

Hazard pictograms

GHS05

GHS07

GHS09

Signal word
Danger

Hazardous component(s) to be indicated on label:
copper dihydroxide

Hazard statement(s)

H302	Harmful if swallowed.
H318	Causes serious eye damage.
H410	Very toxic to aquatic life with long lasting effects.

Hazard statements (EU)

EUH401	To avoid risks to human health and the environment, comply with the instructions for use.

Precautionary statement(s)

P102	Keep out of reach of children.
P270	Do not eat, drink or smoke when using this product.
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310	Immediately call a POISON CENTER/doctor.
P405	Store locked up.
P501	Dispose of contents/container to a facility in accordance with local/regional/national/international regulations.

2.3 Other hazards
No data available.

SECTION 3: Composition/information on ingredients

3.1 Substances
Not applicable. The product is not a substance.

3.2 Mixtures

Chemical characterization
Mixture (preparation)

Hazardous ingredients

No	Substance name		Additional information	
	CAS / EC / Index / REACH no	Classification (EC) 1272/2008 (CLP)	Concentration	%
1	**copper dihydroxide**			
	20427-59-2 243-815-9 029-021-00-3 01-2119969283-29	Acute Tox. 2; H330 Acute Tox. 4; H302 Aquatic Acute 1; H400 Aquatic Chronic 1; H410 Eye Dam. 1; H318	>= 70.00 - < 90.00	wt%
2	**Limestone**			
	1317-65-3 215-279-6 - -	-	>= 5.00 - < 10.00	wt%
3	**sucrose**			
	57-50-1 200-334-9 - -	-	< 2.50	wt%

Full Text for all H-phrases and EUH-phrases: pls. see section 16

No	Note	Specific concentration limits	M-factor (acute)	M-factor (chronic)

1	-	-	M = 10	-

Acute toxicity estimate (ATE) values			
No	**oral**	**dermal**	**inhalative**
1	451 mg/kg bodyweight		0,451 mg/l

SECTION 4: First aid measures

4.1 Description of first aid measures

General information
In case of persisting adverse effects, consult a physician. Remove contaminated clothing and shoes immediately, and launder thoroughly before reusing. Poisonous symptoms can first be observed after several hours, therefore medical observation for at least 48 hours is necessary.

After inhalation
Remove affected person from the immediate area. Ensure supply of fresh air. If unconscious place in recovery position and seek medical advice.

After skin contact
Wash off immediately with soap and water.

After eye contact
Remove contact lenses. Rinse eye thoroughly under running water keeping eyelids wide open and protecting the unaffected eye (at least 10 to 15 minutes). Seek medical assistance.

After ingestion
Call a doctor immediately and show label or packaging. Rinse the mouth thoroughly with water. Never give anything by mouth to an unconscious person.

4.2 Most important symptoms and effects, both acute and delayed
No data available.

4.3 Indication of any immediate medical attention and special treatment needed
No data available.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media
Product itself is non-combustible; adapt fire extinguishing measures to surrounding areas. Carbon dioxide; Extinguishing powder; Water spray jet; Foam

Unsuitable extinguishing media
High power water jet

5.2 Special hazards arising from the substance or mixture
In the event of fire, the following can be released: Carbon dioxide (CO2); Carbon monoxide (CO)

5.3 Advice for firefighters
Fire residues and contaminated firefighting water must be disposed of in accordance with the local regulations. Fire-fighting operations, rescue and clearing work under effect of combustion and smoulder gases just may be done with breathing apparatus. Wear protective clothing.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel
Refer to protective measures listed in sections 7 and 8. Ensure adequate ventilation. Avoid dust formation.

For emergency responders
No data available. Personal protective equipment (PPE) - see Section 8.

6.2 Environmental precautions
Do not discharge into the drains/surface waters/groundwater.

6.3 Methods and material for containment and cleaning up

Collect mechanically. When collected, handle material as described under the section heading "Disposal considerations".

6.4 Reference to other sections
No data available.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling
Avoid the formation and deposition of dust. Provide good ventilation at the work area (local exhaust ventilation, if necessary). Avoid contact with skin and eyes.

General protective and hygiene measures
Do not eat, drink or smoke during work time. Keep away from foodstuffs and beverages. Wash hands before breaks and after work. Do not inhale dust.

Advice on protection against fire and explosion
No special measures necessary.

7.2 Conditions for safe storage, including any incompatibilities

Technical measures and storage conditions
Keep container tightly closed and dry in a cool, well-ventilated place.

Recommended storage temperature

Value	0	-	30	°C

Requirements for storage rooms and vessels
Store product in closed containers.

Incompatible products
Do not store together with foodstuffs.

7.3 Specific end use(s)

Industry solution
Always read the label and product information before use.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational exposure limit values

No	Substance name	CAS no.		EC no.	
1	**copper dihydroxide**	**20427-59-2**		**243-815-9**	
	List of approved workplace exposure limits (WELs) / EH40				
	Copper and compounds (as Cu) dusts and mists				
	WEL short-term (15 min reference period)	2	mg/m³		
	WEL long-term (8-hr TWA reference period)	1	mg/m³		
2	**Limestone**	**1317-65-3**		**215-279-6**	
	List of approved workplace exposure limits (WELs) / EH40				
	Calcium carbonate inhalable dust				
	WEL long-term (8-hr TWA reference period)	10	mg/m³		
	List of approved workplace exposure limits (WELs) / EH40				
	Calcium carbonate respirable dust				
	WEL long-term (8-hr TWA reference period)	4	mg/m³		
3	**sucrose**	**57-50-1**		**200-334-9**	
	List of approved workplace exposure limits (WELs) / EH40				
	Sucrose				
	WEL short-term (15 min reference period)	20	mg/m³		
	WEL long-term (8-hr TWA reference period)	10	mg/m³		
4	**methanol**	**67-56-1**		**200-659-6**	
	2006/15/EC				
	Methanol				
	WEL long-term (8-hr TWA reference period)	260	mg/m³	200	ppm

	Skin resorption / sensibilisation	Skin			
	List of approved workplace exposure limits (WELs) / EH40				
	Methanol				
	WEL short-term (15 min reference period)	333	mg/m³	250	ppm
	WEL long-term (8-hr TWA reference period)	266	mg/m³	200	ppm
	Comments	Sk			

DNEL, DMEL and PNEC values

DNEL values (worker)

No	Substance name			CAS / EC no	
	Route of exposure	**Exposure time**	**Effect**	**Value**	
1	**copper dihydroxide**			**20427-59-2** **243-815-9**	
	dermal			9566.9	mg/kg/day
	with reference to: solid				
	dermal			956.9	mg/kg/day
	with reference to: slurry				
	inhalative			1	mg/m³

PNEC values

No	Substance name		CAS / EC no	
	ecological compartment	**Type**	**Value**	
1	**copper dihydroxide**		**20427-59-2** **243-815-9**	
	water	fresh water	0.0078	mg Cu L-1
	water	fresh water sediment	87	mg Cu kg dwt-1
	water	marine water	0.0052	mg Cu L-1
	water	marine water sediment	676	mg Cu kg dwt-1
	soil	-	65	mg Cu kg dwt-1
	sewage treatment plant	-	0.23	mg Cu L-1

8.2 Exposure controls

Appropriate engineering controls
No data available.

Personal protective equipment

Respiratory protection
If workplace exposure limits are exceeded, a respiration protection approved for this particular job must be worn. In case of dust formation, take appropriate measures for breathing protection in the event that workplace threshold values are not specified.
Respiratory filter (part): P2

Eye / face protection
Tightly fitting safety glasses (EN 166).

Hand protection
In case of intensive contact, wear protective gloves (EN 374). Sufficient protection is given wearing suitable protective gloves checked according to i.e. EN 374, in the event of risk of skin contact with the product. Before use, the protective gloves should be tested in any case for its specific work-station suitability (i.e. mechanical resistance, product compatibility and antistatic properties). Adhere to the manufacturer's instructions and information relating to the use, storage, care and replacement of protective gloves. Protective gloves shall be replaced immediately when physically damaged or worn. Design operations thus to avoid permanent use of protective gloves.

Appropriate Material	nitrile		
Material thickness	>	0.3	mm
Breakthrough time	>	480	min
Appropriate Material	rubber		
Material thickness	>	0.5	mm
Breakthrough time	>	240	min

Other
Normal chemical work clothing.

Environmental exposure controls
No data available.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

State of aggregation
solid

Form
Powder

Colour
light blue

Odour
non specific

pH value	
Value	7.0 - 8.5
Reference temperature	20 °C
Method	CIPAC MT 75.3

Boiling point / boiling range
Not applicable

Melting point/freezing point
not determined

Decomposition temperature
No data available

Flash point	
Method	EEC A9
Comments	non-flammable

Ignition temperature
No data available

Auto-ignition temperature	
Comments	Product is not selfigniting.

Oxidising properties	
not oxidizing	
Method	EEC A17

Explosive properties
Product does not present an explosion hazard.

Flammability
The substance is not flammable.

Lower explosion limit
No data available

Upper explosion limit
No data available

Vapour pressure
Not applicable

Relative vapour density
Not applicable

Relative density
No data available

Density
No data available

Bulk density				
Value	385	-	485	g/l
Method	CIPAC MT 33			
Comments	loose			
Value	575	-	675	g/l
Method	CIPAC MT 33			
Comments	solid			

Solubility in water	
Comments	dispersible

Solubility
No data available

Partition coefficient n-octanol/water (log value)
No data available

Kinematic viscosity
Not applicable

Particle characteristics

9.2 Other information

Other information
No data available.

SECTION 10: Stability and reactivity

10.1 Reactivity
No data available.

10.2 Chemical stability
Stable under recommended storage and handling conditions (See section 7).

10.3 Possibility of hazardous reactions
No data available.

10.4 Conditions to avoid
Heat, naked flames and other ignition sources.

10.5 Incompatible materials
None known.

10.6 Hazardous decomposition products
No hazardous decomposition products known.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute oral toxicity			
No	**Product Name**		
1	**Cupros 50 WP**		
LD50		1750	mg/kg
Species	rat (male)		
Source	Manufacturer		
LD50		1250	mg/kg
Species	rat (female)		
Method	OECD 401		
Source	Manufacturer		

Acute dermal toxicity			
No	**Product Name**		
1	**Cupros 50 WP**		
LD50	>	2000	mg/kg

Species	rats (male/female)
Method	OECD 402
Source	Manufacturer

Acute inhalational toxicity			
No	**Product Name**		
1	**Cupros 50 WP**		
LC50		5.45	mg/l
Duration of exposure		4	h
State of aggregation	Dust/mist		
Species	rat		
Source	Manufacturer		

Skin corrosion/irritation	
No	**Product Name**
1	**Cupros 50 WP**
Species	rabbit
Method	OECD 404
Source	Manufacturer
Evaluation	non-irritant

Serious eye damage/irritation	
No	**Product Name**
1	**Cupros 50 WP**
Species	rabbit
Method	OECD 405
Source	Manufacturer
Evaluation	irritant - risk of strong eye injuries

Respiratory or skin sensitisation	
No	**Product Name**
1	**Cupros 50 WP**
Route of exposure	Skin
Species	guinea pig
Method	OECD 406
Evaluation	Non-sensitizing

Germ cell mutagenicity			
No	**Substance name**	**CAS no.**	**EC no.**
1	**copper dihydroxide**	**20427-59-2**	**243-815-9**
Source	European Union Risk Assessment Report		
Evaluation/classification	Based on available data, the classification criteria are not met.		

Reproduction toxicity			
No	**Substance name**	**CAS no.**	**EC no.**
1	**copper dihydroxide**	**20427-59-2**	**243-815-9**
with reference to	copper compounds		
Source	European Union Risk Assessment Report		
Evaluation/classification	Based on available data, the classification criteria are not met.		

Carcinogenicity			
No	**Substance name**	**CAS no.**	**EC no.**
1	**copper dihydroxide**	**20427-59-2**	**243-815-9**
with reference to	copper compounds		
Source	European Union Risk Assessment Report		
Evaluation/classification	Based on available data, the classification criteria are not met.		

STOT - single exposure			
No	**Substance name**	**CAS no.**	**EC no.**
1	**copper dihydroxide**	**20427-59-2**	**243-815-9**
Source	Manufacturer		
Evaluation/classification	Based on available data, the classification criteria are not met.		

STOT - repeated exposure

No	Substance name	CAS no.	EC no.
1	copper dihydroxide	20427-59-2	243-815-9

Source	Manufacturer
Evaluation/classification	Based on available data, the classification criteria are not met.

Aspiration hazard
No data available

Delayed and immediate effects as well as chronic effects from short and long-term exposure
Inhalation of dusts may irritate the respiratory tract.

11.2 Information on other hazards

Endocrine disrupting properties
No data available.

Other information
No data available.

SECTION 12: Ecological information

12.1 Toxicity

Toxicity to fish (acute)			
No	Product Name		
1	Cupros 50 WP		
LC50		16.5	µg Cu/l
Duration of exposure		96	h
Species	Oncorhynchus mykiss		
Method	OECD 203		
Source	Manufacturer		

Toxicity to fish (chronic)
No data available

Toxicity to Daphnia (acute)			
No	Product Name		
1	Cupros 50 WP		
EC50		2.4	mg/l
Duration of exposure		48	h
Species	Daphnia		
Method	OECD 202		
Source	Manufacturer		

Toxicity to Daphnia (chronic)
No data available

Toxicity to algae (acute)			
No	Product Name		
1	Cupros 50 WP		
EC50		18.94	µg/l
Duration of exposure		72	h
Species	Selenastrum capricornutum		
Method	OECD 201		
Source	Manufacturer		

Toxicity to algae (chronic)
No data available

Bacteria toxicity
No data available

12.2 Persistence and degradability

Biodegradability			
No	Substance name	CAS no.	EC no.
1	copper dihydroxide	20427-59-2	243-815-9

Evaluation	Not applicable for inorganic substances.

12.3 Bioaccumulative potential
No data available.

12.4 Mobility in soil
No data available.

12.5 Results of PBT and vPvB assessment
No data available.

12.6 Endocrine disrupting properties
No data available.

12.7 Other adverse effects
No data available.

12.8 Other information

Other information
Do not discharge product unmonitored into the environment.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product

Waste code	02 01 08*	agrochemical waste containing hazardous substances

Packaging
Residues must be removed from packaging and when emptied completely disposed of in accordance with the regulations for waste removal. Incompletely emptied packaging must be disposed of in the form of disposal specified by the regional disposer.

SECTION 14: Transport information

14.1 Transport ADR/RID/ADN

Class	9
Classification code	M7
Packing group	III
Hazard identification no.	90
UN number	UN3077
Proper shipping name	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.
Technical name	copper dihydroxide
Tunnel restriction code	-
Label	9
Environmentally hazardous substance mark	Symbol "fish and tree"

14.2 Transport IMDG

Class	9
Packing group	III
UN number	UN3077
Proper shipping name	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.
Technical name	copper dihydroxide
EmS	F-A, S-F
Label	9
Marine pollutant mark	Symbol "fish and tree"

14.3 Transport ICAO-TI / IATA

Class	9
Packing group	III
UN number	UN3077
Proper shipping name	Environmentally hazardous substance, solid, n.o.s.
Technical name	copper dihydroxide
Label	9

Environmentally hazardous substance mark	Symbol "fish and tree"

14.4 Other information
No data available.

14.5 Environmental hazards
Information on environmental hazards, if relevant, please see 14.1 - 14.3.

14.6 Special precautions for user
No data available.

14.7 Maritime transport in bulk according to IMO instruments
Not relevant

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU regulations

Regulation (EC) No 1907/2006 (REACH) Annex XIV (List of substances subject to authorisation)
According to the data available and/or specifications supplied by upstream suppliers, this product does not contain any substances considered as substances requiring authorisation as listed on Annex XIV of the REACH regulation (EC) 1907/2006.

REACH candidate list of substances of very high concern (SVHC) for authorisation
According to available data and the information provided by preliminary suppliers, the product does not contain substances that are considered substances meeting the criteria for inclusion in annex XIV (List of Substances Subject to Authorisation) as laid down in Article 57 and article 59 of REACH (EC) 1907/2006.

Regulation (EC) No 1907/2006 (REACH) Annex XVII: RESTRICTIONS ON THE MANUFACTURE, PLACING ON THE MARKET AND USE OF CERTAIN DANGEROUS SUBSTANCES, MIXTURES AND ARTICLES

The product contains following substance(s) that are considered being subject to REACH regulation (EC) 1907/2006 annex XVII.

No	Substance name	CAS no.	EC no.	No
1	copper dihydroxide	20427-59-2	243-815-9	75
2	Limestone	1317-65-3	215-279-6	75

Directive 2012/18/EU on the control of major-accident hazards involving dangerous substances	
This product is subject to Part I of Annex I, risk category:	E1

15.2 Chemical safety assessment
A chemical safety assessment has not been carried out for this mixture.

SECTION 16: Other information

Sources of key data used to compile the data sheet:
Regulation (EC) No 1907/2006 (REACH), 1272/2008 (CLP) as amended in each case.
Directives 2000/39/EC, 2006/15/EC, 2009/161/EU, (EU) 2017/164.
National Threshold Limit Values of the corresponding countries as amended in each case.
Transport regulations according to ADR, RID, IMDG, IATA as amended in each case.
The data sources used to determine physical, toxic and ecotoxic data, are indicated directly in the corresponding section.

Full text of the H- and EUH- phrases drawn up in sections 2 and 3 (provided not already drawn up in these sections)

H330	Fatal if inhaled.
H400	Very toxic to aquatic life.

Creation of the safety data sheet
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This information is based on our present knowledge and experience.
The safety data sheet describes products with a view to safety requirements.
It does not however, constitute a guarantee for any specific product properties and shall not establish a legally valid contractual relationship.

Alterations/supplements:
Alterations to the previous edition are marked in the left-hand margin.

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